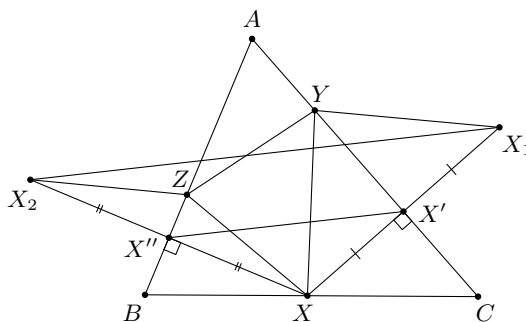


Problem 29. Consider $\triangle ABC$ with $BC = a$, $CA = b$, $AB = c$, and circumradius R . Let X, Y, Z be points on the interior of sides BC, CA and AB , respectively. Prove that

$$(XY + YZ + ZX) \left(\frac{1}{AX} + \frac{1}{BY} + \frac{1}{CZ} \right) \geq \frac{a+b+c}{R}.$$

Solution.



Let $\ell = \frac{XY + YZ + ZX}{2}$. Reflect X about CA to get X_1 , with XX_1 intersecting CA at X' . Reflect X about AB to get X_2 , with XX_2 intersecting AB at X'' . Then

$$2\ell = XY + YZ + ZX = X_1Y + YZ + ZX_2 \geq X_1X_2 = 2X'X'',$$

so $\ell \geq X'X''$.

Now observe that $AX'X''$ is cyclic with AX as diameter, so by the Sine Formula,

$$\frac{X'X''}{\sin A} = \frac{AX'}{\sin \angle AXX'} = AX.$$

Hence

$$\frac{\ell}{AX} \geq \frac{X'X''}{AX} = \sin A.$$

Similarly,

$$\frac{\ell}{BY} \geq \sin B \quad \text{and} \quad \frac{\ell}{CZ} \geq \sin C.$$

Therefore, add and get

$$\ell \left(\frac{1}{AX} + \frac{1}{BY} + \frac{1}{CZ} \right) \geq \sin A + \sin B + \sin C = \frac{a+b+c}{2R},$$

and the result follows. □

Problem 30. Consider $\triangle ABC$ with $BC = a$, $CA = b$, and $AB = c$. Prove that

$$\frac{2bc \cos A}{b+c} < b+c-a < \frac{2bc}{a}.$$

Solution. If $a \leq \max\{b, c\}$, then $\max\{b, c\} < a + \min\{b, c\}$ and so

$$\begin{aligned} b+c-a &= \min\{b, c\} + \max\{b, c\} - a \\ &< \min\{b, c\} + \min\{b, c\} + a - a \\ &= 2\min\{b, c\} \\ &\leq 2\min\{b, c\} \cdot \frac{\max\{b, c\}}{a} \\ &= \frac{2bc}{a}. \end{aligned}$$

If $a > \max\{b, c\}$, then $a < 2\max\{b, c\}$ must hold (otherwise $a \geq b+c$), and so

$$\begin{aligned} b+c-a &= \min\{b, c\} + \max\{b, c\} - a \\ &< \min\{b, c\} + a - a \\ &= \min\{b, c\} \\ &< \min\{b, c\} \cdot \frac{2\max\{b, c\}}{a} \\ &= \frac{2bc}{a}. \end{aligned}$$

Hence right-side inequality holds.

Now observe that

$$(b+c)(b+c-a) - 2bc \cos A = a \left(\frac{2bc}{a} - (b+c-a) \right) > 0$$

which is true because of the right-side inequality. □

Problem 31. Let G , R and r be, respectively, the centroid, the circumradius and inradius of $\triangle ABC$. Prove that

$$2r \leq \frac{AG + BG + CG}{3} \leq R.$$

Solution. We use the following facts:

- $\frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c} = \frac{1}{r}$,
- $AG^2 + BG^2 + CG^2 = 3(R^2 - OG^2) \leq 3R^2$,

where h_a, h_b, h_c are, respectively, the lengths of the heights from A, B, C to BC, CA, AB , and O is the circumcentre of $\triangle ABC$.

For the left-side inequality, note that the distances from G to BC, CA, AB are $\frac{h_a}{3}, \frac{h_b}{3}, \frac{h_c}{3}$, respectively, and so by the Erdős-Mordell Inequality and the Cauchy-Schwarz Inequality, we have

$$\begin{aligned} \frac{AG + BG + CG}{3} &\geq \frac{1}{3} \cdot 2 \left(\frac{h_a}{3} + \frac{h_b}{3} + \frac{h_c}{3} \right) \\ &= \frac{2}{9} \cdot r(h_a + h_b + h_c) \left(\frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c} \right) \\ &\geq \frac{2}{9} \cdot r(1+1+1)^2 \\ &= 2r \end{aligned}$$

For the right-side inequality, we use the Cauchy-Schwarz Inequality, and get

$$\frac{AG + BG + CG}{3} \leq \frac{1}{3} \sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{AG^2 + BG^2 + CG^2} \leq \frac{1}{3} \sqrt{3} \cdot \sqrt{3R^2} = R.$$

□

Problem 32. Given $\triangle ABC$, prove that

$$\sin A \sin B \sin C \left(\frac{1}{\sin A + \sin B} + \frac{1}{\sin B + \sin C} + \frac{1}{\sin C + \sin A} \right) \leq \frac{3}{4} (\cos A + \cos B + \cos C).$$

Solution. We use the following facts:

- $\sin^2 A + \sin^2 B + \sin^2 C = \frac{a^2}{4R^2} + \frac{b^2}{4R^2} + \frac{c^2}{4R^2} = \frac{s^2 - 4Rr - r^2}{2R^2}$,
- $\sin A \sin B + \sin B \sin C + \sin C \sin A = \frac{ab}{4R^2} + \frac{bc}{4R^2} + \frac{ca}{4R^2} = \frac{r^2 + s^2 + 4Rr}{4R^2}$,
- $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$,

where $BC = a, CA = b, AB = c, R$ is the circumradius, r is the inradius and s is the semi-perimeter.

Let L be the left-hand side of the inequality. Then

$$\begin{aligned} L &\leq \sin A \sin B \sin C \left(\frac{1}{2\sqrt{\sin A \sin B}} + \frac{1}{2\sqrt{\sin B \sin C}} + \frac{1}{2\sqrt{\sin C \sin A}} \right) \\ &= \frac{1}{2} \left(\sin A \sqrt{\sin B \sin C} + \sin B \sqrt{\sin C \sin A} + \sin C \sqrt{\sin A \sin B} \right) \\ &\leq \frac{1}{2} \sqrt{\sin^2 A + \sin^2 B + \sin^2 C} \cdot \sqrt{\sin B \sin C + \sin C \sin A + \sin A \sin B} \\ &\leq \frac{1}{2} \sqrt{\frac{(4R^2 + 4Rr + 3r^2) - 4Rr - r^2}{2R^2}} \cdot \sqrt{\frac{r^2 + (4R^2 + 4Rr + 3r^2) + 4Rr}{4R^2}} \\ &= \frac{1}{2} \sqrt{\frac{4R^2 + 2r^2}{2R^2}} \cdot \sqrt{\frac{4R^2 + 8Rr + 4r^2}{4R^2}} \\ &= \frac{1}{2} \sqrt{2 + \left(\frac{r}{R} \right)^2} \cdot \left(1 + \frac{r}{R} \right) \\ &\leq \frac{1}{2} \sqrt{2 + \left(\frac{1}{2} \right)^2} (\cos A + \cos B + \cos C) \\ &= \frac{3}{4} (\cos A + \cos B + \cos C), \end{aligned}$$

where we used the AM-GM Inequality in the first inequality, Cauchy-Schwarz Inequality in the second inequality, Gerretsen's Inequality in the third inequality, and finally, Euler's Inequality in the last inequality. □